

| Basis             | Classification                                     | Prediction                                   |
|-------------------|----------------------------------------------------|----------------------------------------------|
| Definition        | Assigns data into predefined classes or categories | Estimates a continuous numeric value         |
| Output Type       | Discrete (labels/classes)                          | Continuous (numeric values)                  |
| Goal              | Identify class/category of data                    | Forecast future or unknown values            |
| Techniques Used   | Decision Trees, Naive Bayes, KNN, SVM              | Regression (Linear, Polynomial), Time Series |
| Nature of Problem | Categorical problem                                | Numerical problem                            |
| Example           | Classifying emails as spam or not spam             | Predicting house price based on features     |
| Another Example   | Classifying patients as diseased or healthy        | Predicting temperature for next day          |

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| Feature         | Bagging (Bootstrap Aggregating)                                                | Boosting                                                                        |
|-----------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Goal            | Reduce <b>variance</b> (overfitting)                                           | Reduce <b>bias</b> (underfitting)                                               |
| Training        | Base models trained <b>independently</b> in parallel                           | Base models trained <b>sequentially</b> ; each tries to correct previous errors |
| Data sampling   | <b>Bootstrap samples</b> (random draws with replacement) from original dataset | Entire dataset used, but <b>weights</b> of misclassified samples are increased  |
| Weighting       | All models have <b>equal weight</b> in final vote                              | Models have <b>different weights</b> (better models get more say)               |
| Base models     | Usually <b>high-variance, low-bias</b> models (e.g., deep trees)               | Usually <b>weak learners</b> (e.g., shallow trees, stumps)                      |
| Effect on noise | Robust to noisy data                                                           | Can be sensitive to outliers/noise                                              |
| Example         | Random Forest                                                                  | AdaBoost, Gradient Boosting, XGBoost                                            |

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| Aspect                  | BI System                                       | Traditional Reporting                                        |
|-------------------------|-------------------------------------------------|--------------------------------------------------------------|
| <b>Purpose</b>          | Discover insights, support decision making      | Provide static, periodic summaries                           |
| <b>Data freshness</b>   | Real-time or near real-time                     | Batch, often daily/weekly/monthly                            |
| <b>Interactivity</b>    | High (drill-down, slice-dice, what-if analysis) | Low (static reports, dashboards limited)                     |
| <b>User type</b>        | Analysts, managers, data scientists             | Operational staff, compliance                                |
| <b>Query complexity</b> | Ad-hoc, complex, iterative                      | Predefined, fixed SQL queries                                |
| <b>Tools</b>            | Power BI, Tableau, Looker, Qlik                 | Crystal Reports, legacy SQL Server Reporting Services (SSRS) |
| <b>Data model</b>       | Multi-dimensional (star schema, data cubes)     | Simple normalized or flat tables                             |
| <b>Self-service</b>     | Yes (business users explore data)               | No (requires IT to generate reports)                         |
| <b>Focus</b>            | Why did it happen? What will happen?            | What happened? (historical record)                           |